

Hypothesis Testing for Linear Models

From linear restrictions to exact F tests, partial tests, and power

$$H_0 : C\beta = d \quad \implies \quad F = \frac{\text{extra SS}/q}{MSE}$$

Learning objectives

The central question: how do linear restrictions on β become exact finite-sample tests?

ROADMAP

- Write hypotheses in the general linear form $H_0: C\beta = d$ and recognize common special cases.
- Connect testability to estimability, especially when the design matrix is rank deficient.
- Derive restricted least squares and the extra sum of squares associated with q restrictions.
- Use the quadratic-form results from the previous lecture to obtain exact chi-squared, t , and F distributions.
- Interpret global F tests, partial F tests, ANOVA contrasts, confidence regions, and power within one framework.

Bridge from the previous lecture

REVIEW

The quadratic-form and Cochran results already gave us the distributional machinery.

Residual variation

For a rank- p full model,
 $SSE/\sigma^2 \sim \chi^2_{n-p}$.

Extra variation

For nested model spaces, the reduction in SSE is a quadratic form in H_1-H_0 .

Independence

Orthogonal projection pieces are independent under normality.

The consequence

$$\frac{(\text{extra SS})/q}{SSE_F/(n-p)} \sim F_{q,n-p} \quad \text{under } H_0$$

Today we make the hypothesis itself - $C\beta = d$ - the starting point.

01

General linear hypotheses

A single notation covers coefficient tests, contrasts, global tests, and nested-model comparisons.

Normal linear model setup

MODEL

Full-rank model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I), \quad \text{rank}(X) = p$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$$

Residual variance estimate

MSE = SSE/(n-p), with $\text{SSE}/\sigma^2 \sim \chi^2_{n-p}$.

Key independence

$\hat{\beta}$ is independent of SSE under the normal model.

The general linear hypothesis

HYPOTHESIS

Let C be a $q \times p$ matrix of rank q , and let $d \in \mathbb{R}^q$.

General linear hypothesis

$$H_0 : C\beta = d \quad \text{versus} \quad H_1 : C\beta \neq d$$

q restrictions

After redundant rows are removed, $q = \text{rank}(C)$ is the number of independent restrictions being tested.

What is being tested?

Whether the true coefficient vector lies in the subset of parameter space satisfying those restrictions.

Many familiar tests are the same object

EXAMPLES

Question	Hypothesis	C and d
Is β_2 zero?	$H_0: \beta_2 = 0$	$C = (0, 1, 0, \dots), d = 0$
Are β_2 and β_3 equal?	$H_0: \beta_2 - \beta_3 = 0$	$C = (0, 1, -1, 0, \dots), d = 0$
Are all slopes zero?	$H_0: \beta_2 = \dots = \beta_p = 0$	C selects the slope coefficients
Does a set of terms add anything?	$H_0: \text{added coefficients} = 0$	C selects the added block
Does a contrast equal a target?	$H_0: c^T \beta = d$	$C = c^T$

A hypothesis must be testable

ESTIMABILITY

In a full-rank model every linear function of β is estimable. In a rank-deficient model this is no longer true.

Testability condition

Every row c^T of C must satisfy $c \in \mathcal{C}(X^T)$

Equivalent null-space statement

For every $h \in N(X)$, each tested contrast must satisfy $c^T h = 0$.

Interpretation

The hypothesis may only involve coefficient combinations that are uniquely determined by the mean $X\beta$.

Geometric view of H_0 : $C\beta = d$

GEOMETRY

The unrestricted model allows β to vary over all of $\mathbb{R}^p$. The null hypothesis restricts β to an affine subset.

Null parameter set

$$\mathcal{B}_0 = \{\beta \in \mathbb{R}^p : C\beta = d\}$$

Dimension

$$\dim(\mathcal{B}_0) = p - q$$

If $d = 0$

The null set is a linear subspace through the origin.

If $d \neq 0$

The null set is a translated subspace. Restricted least squares still works; we can re-center the problem.

Nonzero d can be reduced to a homogeneous restriction

REPARAMETERIZATION

Choose any β_0 satisfying $C\beta_0 = d$, and write $\beta = \beta_0 + \gamma$.

$$C\beta = d \iff C(\beta_0 + \gamma) = d \iff C\gamma = 0$$

$$Y - X\beta_0 = X\gamma + \varepsilon$$

Why this helps

After subtracting the known offset $X\beta_0$, the null hypothesis becomes a linear subspace problem.

Connection to Lecture 6

This is what lets projection-matrix and Cochran arguments handle general $C\beta = d$, not only zero restrictions.

The null model is a restricted version of the full model

NESTED MODELS

After reparameterization, the null and full mean spaces satisfy

$$\mathcal{M}_0 \subset \mathcal{M}_1 = \mathcal{C}(X)$$

Full fit

Choose the closest point to Y in the full model space $\mathcal{M}_1$.

Restricted fit

Choose the closest point to Y among means satisfying H_0 .

Evidence against H_0

Measure how much residual sum of squares increases when the restrictions are imposed.

Core testing quantity

$$\text{Extra SS} = SSE_R - SSE_F \geq 0$$

02

Restricted least squares and extra sums of squares

The test statistic measures the cost of forcing the fitted model to obey H_0 .

Unrestricted versus restricted least squares

LEAST SQUARES

Unrestricted estimator

$\hat{\beta}$ minimizes $SSE(\beta)$ over all $\beta \in \mathbb{R}^p$.

Restricted estimator

$\hat{\beta}_R$ minimizes $SSE(\beta)$ subject to $C\beta = d$.

$$SSE(\beta) = (Y - X\beta)^T (Y - X\beta)$$

Basic inequality

$$SSE_R \geq SSE_F$$

If the unrestricted fit already satisfies H_0 , the two fits coincide and the extra SS is 0.

Restricted least squares: Lagrange multiplier setup

DERIVATION

$$L(\beta, \lambda) = (Y - X\beta)^T (Y - X\beta) + 2\lambda^T (C\beta - d)$$

First-order conditions:

$$X^T X\beta - X^T Y + C^T \lambda = 0$$

$$C\beta = d$$

Interpretation of λ

The multiplier measures how strongly the SSE objective must be adjusted to enforce the restrictions.

Solving these equations gives the restricted estimator on the next slide.

Restricted least-squares estimator

RESULT

Full-rank formula

$$\hat{\beta}_R = \hat{\beta} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)$$

Start at $\hat{\beta}$

The unrestricted estimator is the natural reference point.

Measure violation

$C\hat{\beta} - d$ tells us how far the unrestricted fit misses the restrictions.

Correct only as needed

The middle matrices convert that violation into the minimum-SSE correction in coefficient space.

$$C\hat{\beta}_R = d$$

What the restriction changes

INTERPRETATION

$$\hat{\beta}_R - \hat{\beta} = -(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)$$

No violation

If $C\hat{\beta} = d$, then $\hat{\beta}_R = \hat{\beta}$.

Small violation

A small discrepancy produces a small change, scaled by coefficient precision.

Poorly estimated direction

Restrictions in high-variance directions can be satisfied with relatively little increase in SSE.

The metric is determined by $X^T X$: not all movements in coefficient space cost the same amount of residual sum of squares.

The increase in SSE has a closed form

EXTRA SS

Extra sum of squares

$$SSE_R - SSE_F = (C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d)$$

Always nonnegative

The matrix inside the quadratic form is positive definite when C has q independent rows.

Zero exactly when

The unrestricted estimator satisfies all q restrictions.

Statistical meaning

The numerator compares the observed restriction error to its covariance geometry.

Under H_0 , the restriction error standardizes to $N(0,I)$

DISTRIBUTION

Because $\hat{\beta}$ is multivariate normal,

$$C\hat{\beta} - d \sim N_q(0, \sigma^2 C(X^T X)^{-1} C^T) \quad \text{under } H_0$$

Therefore its squared Mahalanobis length is chi-squared:

Numerator distribution

$$\frac{SSE_R - SSE_F}{\sigma^2} \sim \chi_q^2$$

This is the quadratic-form theorem from Lecture 6 applied in restriction space.

The same numerator is an extra projection sum of squares

PROJECTION VIEW

For nested linear mean spaces $M_0 \subset M_1$ with projection matrices H_0 and H_1 ,

Extra projection

$$P_{extra} = H_1 - H_0, \quad P_{extra}^T = P_{extra}^2 = P_{extra}$$

$$SSE_0 - SSE_1 = Y^T (H_1 - H_0) Y$$

Connection to restrictions

After reparameterizing $C\beta = d$, the restricted model defines M_0 and the unrestricted model defines M_1 .

Why this matters

Now the entire F test can be read as a statement about orthogonal subspaces and their squared lengths.

Why the numerator has q degrees of freedom

DEGREES OF FREEDOM

Rank identity

$$q = \text{rank}(C) = \text{rank}(H_1 - H_0) = \dim(\mathcal{M}_1) - \dim(\mathcal{M}_0)$$

Parameter view

q counts the independent restrictions imposed by H_0 .

Model-space view

q counts the directions present in the full model but removed under H_0 .

Distribution view

q counts the independent squared $N(0,1)$ coordinates in the numerator.

These are three descriptions of the same degrees of freedom.

Under alternatives, the numerator becomes noncentral

POWER

Let $\delta = C\beta - d$ denote the true violation of the null restrictions.

Noncentrality parameter

$$\lambda = \frac{\delta^T [C(X^T X)^{-1} C^T]^{-1} \delta}{\sigma^2}$$

Alternative distribution

$$\frac{SSE_R - SSE_F}{\sigma^2} \sim \chi_q^2(\lambda)$$

Interpretation of λ

Signal strength in the tested directions, measured relative to noise variance and adjusted for the design.

Power

Larger λ shifts the numerator to the right, increasing the probability of rejecting H_0 .

03

Exact F tests

Cochran's theorem supplies the denominator and the independence needed for finite-sample inference.

The full-model SSE supplies an independent estimate of σ^2

DENOMINATOR

From Lecture 6

$$\frac{SSE_F}{\sigma^2} \sim \chi_{n-p}^2, \quad MSE = \frac{SSE_F}{n-p}, \quad E(MSE) = \sigma^2$$

Why use the full model?

The full-model residual space contains the variation left after every fitted direction in X has been removed.

Residual degrees of freedom

$n-p = \text{rank}(I-H)$ is the dimension of that residual subspace.

The denominator is not an arbitrary variance estimate; it is another projection-based quadratic form.

Why the numerator and denominator are independent

COCHRAN

Argument 1: $\hat{\beta}$ versus residuals

The numerator depends on $\hat{\beta}$. The denominator depends on residuals. Under normality, $\hat{\beta} \perp e$, so the two statistics are independent.

Argument 2: projection matrices

For nested models, $H_1 - H_0$ is orthogonal to $I - H_1$. Cochran's theorem gives independence of the corresponding quadratic forms.

Orthogonality identity

$$(H_1 - H_0)(I - H_1) = 0$$

This independence is the step that turns two chi-squared results into an exact F distribution.

The general linear F test

THEOREM

Exact test statistic

$$F = \frac{(C\hat{\beta} - d)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - d) / q}{MSE}$$

Null distribution

$$H_0 : C\beta = d \implies F \sim F_{q, n-p}$$

Equivalent extra-SS form

$$F = [(SSE_R - SSE_F)/q] / [SSE_F/(n-p)].$$

What makes it exact?

Normality plus orthogonal decomposition - not a large-sample approximation.

Size, critical region, p-value, and power

TESTING THEORY

1

Choose a size α

Fix the desired Type I error probability, such as $\alpha = 0.05$.

2

Compute F

Use the observed restriction violation relative to MSE.

3

Reject for large values

Reject H_0 when $F > F_{\{q,n-p;1-\alpha\}}$.

4

Report the p-value

$p = P(F_{\{q,n-p\}} \geq F_{\text{obs}})$ under H_0 .

Under H_1

$$F \sim F_{q,n-p}(\lambda), \quad \text{Power} = P_{\lambda}(F > F_{q,n-p;1-\alpha})$$

One restriction gives the familiar t test

$$Q = 1$$

For $H_0: c^T \beta = d$, define

t statistic

$$t = \frac{c^T \hat{\beta} - d}{\sqrt{MSE \, c^T (X^T X)^{-1} c}}$$

Null law

$$H_0 \implies t \sim t_{n-p}$$

Numerator

A normal estimable function centered at its null value.

Denominator

Its standard error uses MSE, which is independent of the numerator.

For one restriction, $F = t^2$

EQUIVALENCE

$$F = \frac{(c^T \hat{\beta} - d)^2}{MSE \ c^T (X^T X)^{-1} c}$$

$$t^2 = \left(\frac{c^T \hat{\beta} - d}{\sqrt{MSE \ c^T (X^T X)^{-1} c}} \right)^2$$

Hence

$$F_{1, n-p} = t_{n-p}^2$$

Two-sided t test

Rejecting $|t|$ for being large is exactly the same decision as rejecting $F = t^2$ for being large.

Confidence intervals are inverted hypothesis tests

DUALITY

For an estimable function $\theta = c^T\beta$, a $100(1-\alpha)\%$ confidence interval is

Confidence interval

$$c^T \hat{\beta} \pm t_{n-p, 1-\alpha/2} \sqrt{MSE \, c^T (X^T X)^{-1} c}$$

Testing interpretation

$H_0: c^T\beta = d$ is rejected at level α exactly when d lies outside this interval.

Why this matters

Estimation and testing are not separate theories here; they are two views of the same sampling distribution.

For q restrictions: an ellipsoidal confidence region

CONFIDENCE REGION

Let $\theta = C\beta$. The joint $100(1-\alpha)\%$ confidence region is the set of θ satisfying

Joint region

$$(C\hat{\beta} - \theta)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - \theta) \leq q \text{MSE} F_{q, n-p; 1-\alpha}$$

Geometry

The covariance matrix determines the orientation and axes of the ellipsoid.

Test inversion

$H_0: C\beta = d$ is rejected exactly when d lies outside this joint region.

04

Important special cases

Global regression, nested-model comparisons, and ANOVA contrasts are all instances of $H_0: C\beta = d$.

Global regression F test

SPECIAL CASE

Suppose the model contains an intercept and $p-1$ slope coefficients.

$$H_0 : \beta_2 = \cdots = \beta_p = 0$$

The restricted model is intercept-only, so the extra sum of squares is SSR.

Global F statistic

$$F = \frac{SSR / (p - 1)}{SSE / (n - p)} \sim F_{p-1, n-p} \quad \text{under } H_0$$

Interpretation

Does the full predictor space explain more variation than the intercept direction alone?

Partial F test for nested models

SPECIAL CASE

Let $M_0 \subset M_1$ have ranks $p_0 < p_1$, with $q = p_1 - p_0$ added directions.

$$\text{Extra SS} = SSE_0 - SSE_1$$

Partial F statistic

$$F = \frac{(SSE_0 - SSE_1)/q}{SSE_1/(n - p_1)} \sim F_{q, n-p_1} \quad \text{under } H_0$$

Numerator

Variation explained by the added model directions.

Denominator

Residual variation left after fitting the full model.

One-way ANOVA as a general linear hypothesis

ANOVA

Suppose $\tau = (\mu_1, \dots, \mu_g)^T$ contains the g group means.

Equal-means hypothesis

$$H_0 : \mu_1 = \dots = \mu_g$$

One valid contrast matrix is

Rank

$\text{rank}(C) = g-1$, matching the between-group degrees of freedom.

Same F test

The resulting general linear F test is exactly the usual one-way ANOVA F test.

Worked example: jointly testing two slopes

EXAMPLE

Suppose $\beta = (\beta_0, \beta_1, \beta_2)^T$ and we test $H_0: \beta_1 = \beta_2 = 0$.

$$\hat{\beta} = (2.3, 2.0, -1.0)^T, \quad MSE = 4 \quad C(X^T X)^{-1} C^T = \begin{pmatrix} 0.16 & 0.02 \\ 0.02 & 0.09 \end{pmatrix}$$

$$Q = (2.0, -1.0) \begin{pmatrix} 0.16 & 0.02 \\ 0.02 & 0.09 \end{pmatrix}^{-1} \begin{pmatrix} 2.0 \\ -1.0 \end{pmatrix} = 42.86$$

F statistic

$$F = \frac{Q/2}{4} = 5.36$$

Interpretation

Compare 5.36 to $F_{\{2, n-p\}}$. The test asks whether the two slope directions jointly carry signal after accounting for their covariance.

Likelihood-ratio viewpoint

TESTING THEORY

Under normality, the maximized likelihood is determined by the residual sum of squares.

Likelihood ratio

$$\Lambda = \left(\frac{SSE_F}{SSE_R} \right)^{n/2}, \quad -2 \log \Lambda = n \log \left(\frac{SSE_R}{SSE_F} \right)$$

Same ordering

For fixed n and q , $-2 \log \Lambda$ increases monotonically with the extra-SS F statistic.

Important distinction

The classical F test has an exact finite-sample null distribution; generic likelihood-ratio theory often relies on asymptotics.

In the normal linear model, the geometry gives us more than asymptotic theory.

05

Rank-deficient models

Estimability determines which coefficient hypotheses are meaningful and testable.

When X is rank deficient, β is not unique

ESTIMABILITY

Let $\text{rank}(X) = r < p$. Distinct coefficient vectors can produce the same mean $X\beta$.

$$X\beta_1 = X\beta_2 \quad \text{even when} \quad \beta_1 \neq \beta_2$$

Meaningful hypothesis condition

$H_0 : C\beta = d$ is testable only if every row of C is estimable

Bad hypothesis

Testing a nonestimable coefficient by itself depends on an arbitrary identifying constraint.

Good hypothesis

An estimable contrast has the same value for every β representing the same mean.

General linear F test with a generalized inverse

RANK DEFICIENT

Let G be a generalized inverse of $X^T X$, and let $r = \text{rank}(X)$. For q independent estimable restrictions:

Test statistic

$$F = \frac{(C\hat{\beta} - d)^T (C G C^T)^{-1} (C\hat{\beta} - d) / q}{SSE / (n - r)}$$

Null distribution

$$H_0 \implies F \sim F_{q, n-r}$$

Why this is well defined

When rows of C are estimable, $C\hat{\beta}$ and $C G C^T$ do not depend on the arbitrary generalized inverse representation.

The model-space formulation is often safer

GEOMETRY

Instead of asking about nonunique coefficients, formulate the hypothesis directly in terms of the mean vector $\mu = X\beta$.

Full mean space

$$\mu \in \mathcal{M}_1 = \mathcal{C}(X)$$

Null mean space

$$H_0 : \mu \in \mathcal{M}_0 \subset \mathcal{M}_1$$

$$F = \frac{Y^T (H_1 - H_0) Y / q}{Y^T (I - H_1) Y / (n - r_1)}$$

Advantage

Projection matrices H_0 and H_1 are unique even when coefficient parameterizations are not.

Common mistakes to avoid

PITFALLS

1. Testing a nonestimable coefficient

In a rank-deficient model, not every β_j represents an identified quantity.

2. Using the wrong denominator df

The residual df are $n - \text{rank}(X_{\text{full}})$, not automatically $n - p$ when X is rank deficient.

3. Forgetting dependence

A ratio of two chi-squared quantities is F only when the numerator and denominator are independent.

4. Counting written restrictions instead of rank

Degrees of freedom equal $\text{rank}(C)$, after redundant restrictions are removed.

A practical recipe for testing linear hypotheses

WORKFLOW

- 1 State** Write H_0 as $C\beta = d$.
- 2 Check** Verify the rows of C are estimable if X is rank deficient.
- 3 Count** Set $q = \text{rank}(C)$, removing redundant restrictions.
- 4 Fit** Obtain the full-model $\hat{\beta}$, SSE, and MSE.
- 5 Compute** Use either the coefficient-form quadratic statistic or the extra sum of squares.
- 6 Reference** Compare to $F_{\{q, \text{residual df}\}}$ and report the p-value.
- 7 Interpret** Translate the rejection decision back to the scientific restriction, not just coefficients.

Concept check

PRACTICE

- If C has four rows but $\text{rank}(C)=3$, what are the numerator degrees of freedom?
- Why is $H_0: \beta_2 = 0$ potentially meaningless in a rank-deficient model?
- For nested models with ranks 5 and 8 and $n=50$, what are the degrees of freedom of the partial F statistic?
- Under H_1 , what determines the noncentrality parameter of the F distribution?
- Why can the general linear F test be derived both from $\hat{\beta}$ and from projection matrices?
- For $q=1$, how are the t and F tests related?

Takeaways

SUMMARY

Hypothesis

Most classical linear-model tests can be written $H_0: C\beta = d$.

Estimability

In rank-deficient models, only hypotheses involving estimable functions are meaningful.

Extra SS

Imposing H_0 increases SSE; that increase is a quadratic form with q degrees of freedom.

Cochran

Orthogonality gives independence between the tested variation and residual variation.

F test

Dividing the two independent mean squares yields the exact F test.

Power

Under alternatives the numerator, and hence F, becomes noncentral.

linear restriction + normality + orthogonal geometry $\implies$ exact finite-sample test